Heterogeneous Households and Aggregate Dynamics: Analytical Moment Recursion[†]

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Abstract

I develop an analytical framework for heterogeneous-household economies with idiosyncratic productivity and return risk. I prove the existence of a moment-recursive competitive equilibrium (MRCE) in which aggregate capital and two distributional covariances follow a closed-form matrix recursion. The equilibrium holds regardless of household beliefs, accommodating non-stationary growth paths. The closed-form structure enables exact general equilibrium inversion: given U.S. data on output, capital, and hours (1948–2024), I recover structural shocks period by period. The inversion reveals state-dependent amplification from distributional covariances and a sorting accelerator—TFP growth disproportionately benefits productive-wealthy households, providing a structural mechanism for rising inequality.

Keywords: Heterogeneous agent, business cycle, moment-recursive.

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1 Introduction

The distribution of household income, wealth, and portfolio returns shapes aggregate output, employment, and asset prices (Kaplan et al., 2018). Yet the Bewley–Huggett–Aiyagari framework requires tracking the entire wealth distribution—an infinite-dimensional state variable. Krusell and Smith (1998) showed numerically that mean wealth nearly suffices for forecasting prices, but this approximation lacks an analytical foundation and breaks down with richer heterogeneity. This paper closes the gap. I construct a model in which three summary statistics—aggregate capital and two wealth-characteristic covariances—exactly characterize aggregate dynamics via a closed-form matrix recursion. I call the resulting object a *moment-recursive competitive equilibrium* (MRCE).

I make three contributions. First, I derive closed-form policies for consumption, saving, and labor supply under incomplete markets with idiosyncratic productivity and return risk. These policies yield a unique MRCE with law of motion $\mathcal{M}' = \beta\,\widetilde{\mathcal{G}}\,\mathcal{M} + [T, 0, 0]'$, where $\mathcal{M} = [K,\, \mathrm{cov}(z, a),\, \mathrm{cov}(e, a)]$ and $\widetilde{\mathcal{G}}$ depends on current prices. The saving rate is increasing and concave in wealth—consistent with Dynan et al. (2004)—and the borrowing constraint is never binding. Shutting down return heterogeneity nests Krusell–Smith; shutting down all heterogeneity recovers Brock and Mirman (1972). With return risk alone, the wealth distribution drops out entirely as in Angeletos (2007); adding productivity heterogeneity restores the distribution's role through exactly two covariances.

Second, I show the equilibrium is invariant to household beliefs about aggregate dynamics: under log utility, income and substitution effects of future uncertainty cancel exactly, so policies depend only on current prices. This eliminates the fixed-point problem over beliefs, making the MRCE a belief-invariant recursive competitive equilibrium (BRCE). Because policies depend only on current prices, the equilibrium accommodates non-stationary aggregate paths without stationarity or detrending assumptions. This property is essential for studying how TFP growth—as opposed to TFP levels—drives inequality. In the conventional approach, comparing two stationary equilibria at different TFP levels confounds the effect of growth with infinite-horizon adjustment under a permanently constant technology; agents in each steady state do not expect further growth, so the exercise is silent on

how the process of growth itself reshapes the distribution. Computing equilibrium along a non-stationary growth path requires specifying household expectations over the entire future price sequence—a formidable fixed-point problem in heterogeneous-agent economies. The BRCE sidesteps both difficulties: one can feed any TFP path—trending, cyclical, or both—into the moment recursion and obtain valid equilibrium dynamics without imposing stationarity, balanced growth, or rational expectations about the aggregate path.

Third, the closed-form structure enables an *exact general equilibrium inversion* and *structural counterfactual analysis*. Given U.S. data on output, capital, and hours (1948–2024), I recover TFP, time-varying patience, and wealth-leisure complementarity that exactly rationalize the data period by period. Because the BRCE does not require stationarity, the inversion operates directly on data that exhibit both secular growth and business cycle fluctuations—no detrending or balanced-growth assumption is needed. Because these are structural shocks in a model with distributional dynamics, they support counterfactual decompositions—isolating each shock's role in inequality growth and quantifying how amplification depends on the current wealth distribution. This extends the business-cycle accounting of Chari et al. (2007) to a heterogeneous-agent setting.

The inversion yields several substantive findings. The most distinctive is a "sorting accelerator": TFP growth raises wages and returns disproportionately for productive-wealthy households, generating 168 percent of the observed growth in wealth-productivity sorting since 1948 and providing a structural explanation for the concentration patterns documented by Piketty (2014). As sorting accumulates, it creates state-dependent amplification: a 2 percent TFP shock produces 32 percent larger output impact and 64 percent larger cumulative losses from the high-inequality 2022 state than from 1952, so that macroeconomic fragility grows with wealth concentration. TFP captures the output trend but explains only 20 percent of cyclical volatility; the recovered discount factor is strongly countercyclical ($\mathrm{corr}(\beta_t, \log Y_t) = -0.93$) and acts as a powerful stabilizer, as removing it raises output volatility by 50 percent. The model also implies a theoretical risk premium from return heterogeneity (Constantinides and Duffie, 1996), though it is quantitatively negligible at realistic return dispersion ($\sigma_e = 0.01$); the main consequence of return heterogeneity operates through the amplification channel rather than the pricing channel.

Related literature. Bewley (1986), Huggett (1993), and Aiyagari (1994) developed the foundational heterogeneous-agent framework. Krusell and Smith (1998) introduce aggregate risk and show numerically that mean wealth nearly suffices for forecasting prices; the MRCE provides an exact analytical counterpart that clarifies when and why this approximation succeeds. Computational advances for solving heterogeneous-agent models include Reiter (2009), Winberry (2018), and Ahn et al. (2018); the MRCE bypasses these methods entirely by delivering closed-form dynamics. Kaplan et al. (2018) and Auclert et al. (2021) develop heterogeneous-agent New Keynesian models; the sequence-space Jacobian (SSJ) of Auclert et al. (2021) computes linearized impulse responses for general economies, while the MRCE delivers exact nonlinear dynamics for a specific class. The two are complementary: SSJ applies to richer models at the cost of linearization; the MRCE provides exact state-dependent amplification within its scope.

The analytical structure builds on exact aggregation results. Brock and Mirman (1972) solve the stochastic growth model under log utility; the MRCE nests their solution. Angeletos (2007) shows that uninsured return risk admits exact aggregation in which the wealth distribution is irrelevant for aggregate dynamics; the MRCE adds productivity heterogeneity, which brings the distribution back through covariances that evolve via an explicit recursion. Bilbiie (2020) and Ravn and Sterk (2021) develop parsimonious heterogeneous-agent models with analytical aggregate dynamics; this paper uses a continuum of agents with rich return heterogeneity. The belief-invariance result connects to Menzio and Shi (2010) on block-recursive equilibria and Gabaix (2020) on cognitive discounting: the BRCE shows invariance to beliefs about aggregate dynamics themselves, arising endogenously from the income-substitution offset under log utility. Alvarez and Lippi (2022) develop analytical impulse responses using eigenvalue methods in menu-cost models; the moment recursion plays an analogous role for heterogeneous households.

The state-dependent amplification results contribute to the nonlinear business cycle literature: Storesletten et al. (2004) document countercyclical cross-sectional risk, Bloom et al. (2018) study uncertainty-driven state dependence, and Berger and Vavra (2015) analyze wealth heterogeneity in consumption dynamics. The asset pricing results relate to idiosyncratic risk and equilibrium returns (Constantinides and Duffie, 1996; Schmidt, 2016); growing

evidence on heterogeneous portfolio returns (Fagereng et al., 2020; Bach et al., 2020) motivates the model's return channel.

The remainder of the paper proceeds as follows. Section 2 sets up the model environment. Section 3 derives the MRCE, establishes belief invariance, and draws out its macroeconomic and asset pricing implications. Section 4 implements the exact inversion on U.S. data and evaluates the model's quantitative predictions. Section 5 extends the framework to CRRA-GHH and Epstein-Zin preferences. Section 6 concludes. All proofs and supplementary results are collected in the Online Appendix.

2 Model

The economy is populated by a continuum of infinitely lived households who face idiosyncratic labor productivity and portfolio return risk and choose consumption, saving, and labor supply each period.

2.1 States and shocks

Each household's individual state is $x = [a, z, e]$: wealth a (endogenous), idiosyncratic labor productivity z (exogenous), and an idiosyncratic factor e that scales the household's return on capital (exogenous). All three dimensions display substantial cross-sectional variation in the data (Kuhn et al., 2020; Storesletten et al., 2004; Fagereng et al., 2020).

The idiosyncratic processes follow stationary AR(1) dynamics in levels:

$$z' = \rho_z\, z + \varepsilon'_z, \quad \varepsilon'_z \sim \mathcal{N}(0,\, \sigma^2_{\varepsilon_z}), \quad \sigma^2_{\varepsilon_z} = \sigma^2_z(1-\rho^2_z), \tag{1}$$

$$e' = \rho_e\, e + \varepsilon'_e, \quad \varepsilon'_e \sim \mathcal{N}(0,\, \sigma^2_{\varepsilon_e}), \quad \sigma^2_{\varepsilon_e} = \sigma^2_e(1-\rho^2_e), \tag{2}$$

where σ^2_z and σ^2_e denote the unconditional variances and $\varepsilon'_z, \varepsilon'_e$ are mutually independent innovations. Linearity in levels is essential: it ensures that the cross-sectional covariances $\mathrm{cov}(z, a)$ and $\mathrm{cov}(e, a)$ evolve as a closed system, enabling the moment recursion of Section 3. In the quantitative analysis, each process is discretized using the Tauchen (1986) method.

The aggregate state is $X = [\Phi, A, B, C]$, where Φ is the cross-sectional joint distribution

of individual states, A is aggregate TFP, B governs the dispersion of capital returns, and C governs the dispersion of labor productivity. All three exogenous processes follow Markov chains. In the quantitative application (Section 4), only A varies over time; B and C are held at steady state. The framework nevertheless accommodates time-variation in all three, allowing the structure of inequality itself to shift—for instance, a widening of the return distribution during financial crises or a compression of the wage distribution following skill-biased technological change.

2.2 Preferences

Household preferences take a Stone-Geary form with endogenous labor supply:

$$u(c, n, a, z; X) = \log(c - g(x; X)) + \theta \log((\kappa + \xi a) - n), \tag{3}$$

where c is consumption, n is labor supply, and $\theta > 0$ governs the relative weight on leisure.

The subsistence consumption level $g(x; X) = \kappa w(X)(1 - \tau)z$ is proportional to after-tax potential earnings, capturing work-related expenses—commuting, child care, professional attire—that scale with take-home pay.

The effective time endowment $\kappa + \xi a$ increases in wealth: $\kappa \in (0, 1)$ is the base endowment, and $\xi > 0$ captures the degree to which wealth buys usable time through time-saving services or home production (Aguiar and Hurst, 2007). This wealth-leisure complementarity is the primary link between the wealth distribution and aggregate labor supply.

The three parameters (θ, κ, ξ) are calibrated to match average hours, the saving-to-labor-income ratio, and the wealth Gini. Extensions to GHH and Epstein-Zin preferences appear in Section 5.

2.3 Budget constraint and the household problem

The recursive formulation of the household's problem is:

$$v(x;X) = \max_{c,a',n} \quad u(c,n,a,z;X) + \beta \mathbb{E} v(x';X') \tag{4}$$

$$\text{s.t.} \quad c + a' = w(X)(1-\tau)zn + (1 + r(X)(1-\tau)e)a + T(X) \tag{5}$$

$$\Phi' = \Gamma_\Phi(X) \tag{6}$$

$$a' \geq 0, \quad n \geq 0 \tag{7}$$

Consumption plus saving equals after-tax labor income $w(X)(1-\tau)zn$, after-tax capital income $(1+r(X)(1-\tau)e)a$, and a lump-sum transfer $T(X)$. The wage $w(X)$ is per efficiency unit of labor and $r(X)$ is the aggregate return on capital. The household-specific factor e scales this return: $e > 1$ means an above-average return, $e < 1$ a below-average one, and $\mathbb{E}[e] = 1$, so $r(X)$ is the economy-wide average return.

The household discounts the future at rate $\beta \in (0,1)$, faces a borrowing constraint $a' \geq 0$, and takes the law of motion Γ_Φ for the aggregate distribution as given. As shown below, this constraint is non-binding in equilibrium—a consequence of the additive transfer $T(X)$ and the multiplicative structure of the saving policy.

2.4 Production

A representative firm operates a constant-returns-to-scale production technology:

$$Y = AK^\alpha L^{1-\alpha}, \tag{8}$$

where $K = \int a \, d\Phi$ is the aggregate capital stock, $L = \int zn(x;X) \, d\Phi$ is aggregate effective labor, and $\alpha \in (0,1)$ is the capital share. The firm's first-order conditions pin down factor prices:

$$r(X) = \alpha A K^{\alpha-1} L^{1-\alpha} - \delta, \quad w(X) = (1-\alpha) A K^\alpha L^{-\alpha}, \tag{9}$$

where δ is the depreciation rate. Return heterogeneity redistributes income across households without affecting total output. Because factor prices depend on K/L, any distributional force that shifts labor supply feeds back into prices.

2.5 Government policy

The government levies a proportional income tax at rate τ on both capital and labor income, and redistributes a fraction η of tax revenue as a lump-sum transfer to all households:

$$T(X) = \eta \cdot Y(X) \cdot \tau, \tag{10}$$

where $Y(X) = AK^{\alpha}L^{1-\alpha}$ is aggregate output. The remaining revenue $(1-\eta) \cdot Y \cdot \tau$ finances government consumption, which does not enter household utility or the production function.

The additive transfer T plays a critical theoretical role: it provides the intercept in the Kesten process governing individual wealth dynamics, ensuring a well-defined stationary wealth distribution with a Pareto tail (Kesten, 1973). The parameter η controls redistribution: a higher η compresses the distribution, while a lower η allows greater dispersion and a thicker tail.

2.6 Recursive competitive equilibrium

Definition 1 (Recursive competitive equilibrium (RCE))**.**
A collection $(g_c, g_a, g_n, v, \Gamma_\Phi, w, r)$ is a recursive competitive equilibrium if:

1. *g_c, g_a, g_n, v solve the household's problem given Γ_Φ.*
2. *K, L satisfy the production sector's optimality conditions.*
3. *Factor markets clear.*
4. *Γ_Φ satisfies $\Gamma_\Phi(X) = \Phi'$, where Φ' is the future distribution of individual states constructed by applying the intertemporal policy function g_a and the exogenous idiosyncratic state processes to Φ.*

The difficulty is that Φ is infinite-dimensional. The next section shows that optimal household policies depend on Φ only through a three-dimensional vector of moments, and that these moments follow an explicit matrix recursion—collapsing the infinite-dimensional state to a finite system.

3 Moment-Recursive Competitive Equilibrium

3.1 Analytical household policies

The household's optimal consumption, saving, and labor supply all have explicit solutions. These analytical policies are the foundation of all subsequent results: they allow the infinite-dimensional distribution to be replaced by a finite vector of moments and make the equilibrium law of motion an explicit matrix recursion rather than an implicitly defined fixed point.

Proposition 1 (Analytical household-level decisions)**.**
The optimal consumption, saving, and labor supply policies are:

$$c(x;X) = \frac{1-\beta}{1+\theta}(1+r(X)(1-\tau)e)a + \left(\frac{1-\beta}{1+\theta}a\xi + \kappa\right) w(X)(1-\tau)z \tag{11}$$

$$a'(x;X) = \beta a\left((1+r(X)(1-\tau)e) + w(X)(1-\tau)z\xi\right) + T(X) \tag{12}$$

$$n(x;X) = \max\left\{\kappa(1+\theta) + \xi a - \frac{\theta\, c(x;X)}{w(X)(1-\tau)z},\ 0\right\} \tag{13}$$

Proof. The budget constraint can be rewritten as:

$$(1+\theta)(c - \kappa z w(X)(1-\tau)) + a' = \left((1+r(X)(1-\tau)e) + w(X)(1-\tau)z\xi\right) a + T(X). \tag{14}$$

The transfer $T(X)$ is identical across households and therefore enters savings as an additive intercept. The result of Brock and Mirman (1972) immediately applies to the multiplicative component. ■

The consumption function is linear in wealth, but the marginal propensity to consume differs across income sources—higher out of capital income than out of labor income. The

saving function $a' = \beta a(\cdot) + T(X)$ separates into a multiplicative component, which enables analytical aggregation, and an additive transfer, which ensures a stationary wealth distribution.

3.2 The saving rate

The saving rate can be characterized analytically, connecting the model directly to empirical evidence on saving behavior.

Proposition 2 (Analytical saving rate)**.**
The saving rate $s(x;X) = \frac{a'(x;X)-a}{y(x;X)}$*, where* $y(x;X) = w(X)(1-\tau)zn(x;X) + r(X)(1-\tau)ea + T(X)$ *is after-tax income plus transfer, satisfies:*

$$s(x;X) = \frac{a[\beta(1 + r(X)(1-\tau)e + w(X)(1-\tau)z\xi) - 1] + T(X)}{w(X)(1-\tau)z\kappa\theta + a\left[\frac{w(X)(1-\tau)z\xi(1+\theta\beta)}{1+\theta} - \frac{\theta(1-\beta)(1+r(X)(1-\tau)e)}{1+\theta} + r(X)(1-\tau)e\right] + T(X)}, \tag{15}$$

with the following properties:

(i) $\lim_{a\to\infty} s(x;X) = \frac{\beta(1+r(X)(1-\tau)e+w(X)(1-\tau)z\xi)-1}{\frac{w(X)(1-\tau)z\xi(1+\theta\beta)}{1+\theta} - \frac{\theta(1-\beta)(1+r(X)(1-\tau)e)}{1+\theta} + r(X)(1-\tau)e}$

(ii) $\lim_{a\to 0} s(x;X) > 0$ *due to the transfer* $T(X) > 0$

(iii) $\frac{\partial s}{\partial a} > 0$ *and* $\frac{\partial^2 s}{\partial a^2} < 0$.

Proof. See Appendix A.2 of the Online Appendix. ■

The saving rate is positive even at zero wealth (due to the transfer), increases in wealth, and is concave. Richer households save a larger fraction of their income, but the marginal increase in the saving rate diminishes—precisely the pattern documented by Dynan et al. (2004) in the Consumer Expenditure Survey and Survey of Consumer Finances.

The concave saving rate generates mean reversion in relative wealth positions: the subsistence floor absorbs a diminishing share of total resources as wealth increases, so the saving rate converges to a constant in the limit.

3.3 Non-binding borrowing constraint

An important consequence of the policy functions is that the borrowing constraint is always slack:

Corollary 1 (Non-binding borrowing constraint)**.**
The borrowing limit $a' \geq 0$ is always non-binding.

Proof. Given $a'(x; X) = \beta a((1 + r(X)(1 - \tau)e) + w(X)(1 - \tau)z\xi) + T(X) \geq T(X) > 0$, the result is immediate. ■

The non-binding constraint is critical: it keeps policy functions globally smooth and linear, enabling analytical aggregation. This is not merely a technical convenience but a substantive scope condition. The MRCE characterizes economies in which transfers are sufficiently large—relative to borrowing needs—that the constraint never binds. In economies where binding constraints generate kinks in policy functions, the moment recursion breaks down and numerical methods remain necessary. The relevant empirical question is therefore whether the household sector's effective transfer-to-wealth ratio is large enough that the constraint is slack for virtually all agents, a condition broadly consistent with the U.S. economy given Social Security, Medicare, and other social insurance programs.

3.4 Existence and characterization

Definition 2 (Moment-recursive competitive equilibrium (MRCE))**.**
A collection $(g_c, g_a, g_n, v, \Gamma_M, K, L, w, r, \mathcal{M})$ is a moment-recursive competitive equilibrium if:

1. *g_c, g_a, g_n, v solve the household's problem given Γ_M.*
2. *K, L satisfy the production sector's optimality conditions.*
3. *Factor markets clear.*
4. *Γ_M satisfies $\Gamma_M(\mathcal{M}(\Phi), A) = \mathcal{M}(\Phi')$, where $\mathcal{M}$ extracts the sufficient moments from Φ, and Φ' is the distribution of individual states induced by g_a and the exogenous processes.*

The key departure from the RCE is that Γ_M maps a *finite* moment vector forward, rather than the entire distribution. The sufficient statistics turn out to be aggregate capital K and two distributional covariances. The productivity-wealth covariance $\text{cov}(z, a)$ measures the alignment between earnings ability and asset holdings: when it is high, the best earners are also the richest savers. The return-wealth covariance $\text{cov}(e, a)$ measures the alignment between portfolio performance and wealth: when it is high, households earning above-average returns hold a disproportionate share of capital.

Proposition 3 (Unique existence of the MRCE)**.**
The MRCE uniquely exists with $\mathcal{M}(\Phi) = [K, cov(z, a), cov(e, a)]$, and the law of motion Γ_M is as follows. Define after-tax prices $\tilde{r} = r(\mathcal{M}, A)(1 - \tau)$ and $\tilde{w} = w(\mathcal{M}, A)(1 - \tau)$, and the transfer $T = \eta \cdot Y \cdot \tau$. Then:

$$\Gamma_M(\mathcal{M}(\Phi), A) = \begin{bmatrix} K' \\ cov(z', a') \\ cov(e', a') \end{bmatrix} = \beta \widetilde{\mathcal{G}} \begin{bmatrix} K \\ cov(z, a) \\ cov(e, a) \end{bmatrix} + \begin{bmatrix} T \\ 0 \\ 0 \end{bmatrix} \tag{16}$$

where the transition matrix is

$$\widetilde{\mathcal{G}} = \begin{bmatrix} 1 + \tilde{r} + \tilde{w}\xi & \tilde{w}\xi & \tilde{r} \\ \phi_z \rho_z \tilde{w}\xi\sigma_z^2 & \phi_z \rho_z (1 + \tilde{r} + \tilde{w}\xi) & 0 \\ \phi_e \rho_e \tilde{r}\sigma_e^2 & 0 & \phi_e \rho_e (1 + \tilde{r} + \tilde{w}\xi) \end{bmatrix} \tag{17}$$

where prices satisfy:

$$r(\mathcal{M}, A) = \alpha A \left(\frac{L}{K}\right)^{1-\alpha} - \delta \tag{18}$$

$$w(\mathcal{M}, A) = (1 - \alpha) A \left(\frac{K}{L}\right)^{\alpha} \tag{19}$$

$$L(\mathcal{M}, A) = \kappa + \frac{\xi(1 + \theta\beta)(K + cov(z, a))}{1 + \theta} - \frac{\theta(1 - \beta)}{(1 + \theta)\tilde{w}} [K(1 + \tilde{r}) + cov(e, a)\tilde{r}] \tag{20}$$

The transfer T is identical across households and therefore enters only the K' equation. It drops out of the covariance equations because $cov(z', T) = cov(e', T) = 0$.

Proof. See Appendix A.3 of the Online Appendix. ■

This proposition is the main result of the paper. The state vector $\mathcal{M} = [K, \text{cov}(z, a), \text{cov}(e, a)]$ comprises aggregate capital and two distributional covariances that measure the alignment between wealth and idiosyncratic characteristics. Its law of motion is a non-homogeneous linear recursion in which the off-diagonal entries of $\widetilde{\mathcal{G}}$ capture cross-moment feedback.

Two features of $\widetilde{\mathcal{G}}$ deserve emphasis. First, the $(1, 3)$ entry $\tilde{r}$ shows that return-wealth sorting raises future capital, while the $(3, 1)$ entry $\phi_e \rho_e \tilde{r} \sigma_e^2$ shows that higher capital strengthens return-wealth sorting—a self-reinforcing loop. Second, the $(2, 3)$ and $(3, 2)$ entries are zero because z and e are independent: the two sorting mechanisms interact only through aggregate capital. The signal-to-noise ratios $\phi_z \equiv \sigma_z^2/(\sigma_z^2 + \sigma_{\varepsilon_z}^2)$ and $\phi_e \equiv \sigma_e^2/(\sigma_e^2 + \sigma_{\varepsilon_e}^2)$ govern the speed of distributional adjustment, where $\sigma_{\varepsilon_z}^2$ and $\sigma_{\varepsilon_e}^2$ are the innovation variances of the idiosyncratic processes. A detailed interpretation appears in Appendix B.1–B.2.

3.5 Ergodicity

Corollary 2 (Recursivity of MRCE and Kesten stationarity)**.**
The law of motion $\mathcal{M}' = \beta\widetilde{\mathcal{G}}\mathcal{M} + [T, 0, 0]'$ is a non-homogeneous multivariate Kesten process with both multiplicative ($\beta\widetilde{\mathcal{G}}$) and additive ($T$) components. When the spectral radius $\rho(\beta\widetilde{\mathcal{G}}) < 1$, the system admits a unique stationary solution

$$\mathcal{M}^* = (I - \beta\widetilde{\mathcal{G}})^{-1} \begin{bmatrix} T \\ 0 \\ 0 \end{bmatrix}. \tag{21}$$

The additive transfer intercept $T > 0$ is essential: without it, the purely multiplicative system $\mathcal{M}' = \beta\widetilde{\mathcal{G}}\mathcal{M}$ has no well-defined stationary distribution, as the Kesten process requires both multiplicative and additive components to generate ergodic dynamics with Pareto tails in the wealth distribution (Kesten, 1973).

Proof. See Appendix A.4 of the Online Appendix. ■

When the spectral radius condition holds, the economy converges to a unique stationary state $\mathcal{M}^*$, which serves as the starting point for business cycle analysis. The spectral radius of $\beta\widetilde{\mathcal{G}}$ governs the speed of convergence. As shown in Section 4, near-unit-root behavior in

this spectral radius generates slow mean reversion and amplifies the propagation of aggregate shocks.

3.6 Special cases

The MRCE nests several benchmark models, clarifying what each source of heterogeneity contributes to aggregate dynamics.

Corollary 3 (Krusell and Smith (1998) analogue)**.**
Suppose return heterogeneity is muted: $e = 1$. Then, the MRCE uniquely exists with $\mathcal{M}(\Phi) = [K, cov(z, a)]$, and the law of motion is:

$$\Gamma_M(\mathcal{M}(\Phi), A) = \begin{bmatrix} K' \\ cov(z', a') \end{bmatrix} = \begin{bmatrix} \beta\left[(1 + \tilde{r} + \tilde{w}\xi)K + \tilde{w}\xi\, cov(z, a)\right] + T \\ \phi_z \rho_z \beta\left[(1 + \tilde{r} + \tilde{w}\xi)\, cov(z, a) + \tilde{w}\xi\sigma_z^2 K\right] \end{bmatrix} \tag{22}$$

Proof. See Appendix A.5 of the Online Appendix. ■

Without return heterogeneity, the state reduces to $[K, \mathrm{cov}(z, a)]$—the analytical counterpart of the Krusell and Smith (1998) economy. Aggregate capital alone is sufficient if and only if $\mathrm{cov}(z, a) = 0$; the magnitude of this covariance determines when approximate aggregation succeeds and when it fails. This yields a sharp analytical prediction: approximate aggregation holds when $\mathrm{cov}(z, a)$ is small relative to K, which occurs when idiosyncratic productivity shocks are weakly persistent or when wealth mobility is high. The MRCE thus provides exact conditions under which the Krusell and Smith (1998) finding—that mean capital nearly suffices for forecasting—holds as a theorem rather than a numerical observation. Section 4 confirms this quantitatively: the representative-agent channel accounts for 98.3 percent of mean investment. Further discussion appears in Appendix B.3.

Corollary 4 (Representative-agent analogue)**.**
Suppose all heterogeneity is muted: $e = 1$ and $z = 1$. Then, the MRCE uniquely exists with $\mathcal{M}(\Phi) = K$, and the law of motion is:

$$K' = \beta\left[(1 + \tilde{r} + \tilde{w}\xi)K\right] + T, \tag{23}$$

which leads to the aggregate consumption function:

$$C(\mathcal{M}, A) = \frac{1-\beta}{1+\theta}(1 + \tilde{r} + \tilde{w}\xi)K + \kappa\tilde{w}. \tag{24}$$

Proof. See Appendix A.5 of the Online Appendix. ■

With all heterogeneity eliminated, both covariances vanish and the model reduces to the stochastic growth model with log preferences and endogenous labor supply (Brock and Mirman, 1972). Every departure of the full MRCE from this benchmark is therefore attributable to sorting between wealth and idiosyncratic characteristics.

3.7 Belief invariance

A distinctive feature of the model is that the equilibrium is invariant to household beliefs about aggregate dynamics, eliminating the fixed-point problem over beliefs that dominates computation in standard heterogeneous-agent models.

Definition 3 (Belief-invariant recursive competitive equilibrium (BRCE))**.**
A collection $(g_c, g_a, g_n, v, \Gamma_\Phi, K, L, w, r)$ *is a belief-invariant recursive competitive equilibrium if:*

1. g_c, g_a, g_n, v *solve the household's problem given arbitrary beliefs about aggregate Markov laws of motion for both endogenous and exogenous states.*
2. K, L *satisfy the production sector's optimality conditions.*
3. *Factor markets clear.*
4. Γ_Φ *satisfies* $\Gamma_\Phi(X) = \Phi'$*, where* Φ' *is the future distribution of individual states constructed by applying the policy function* g_a *and the exogenous idiosyncratic process to* Φ*.*

Theorem 1 (Belief invariance)**.**
In the BRCE, the law of motion Γ_Φ *is unaffected by households' expectations about future aggregate states.*

Proof. See Appendix A.6 of the Online Appendix. ■

The saving policy $a'(x; X) = \beta a((1+r(X)(1-\tau)e)+w(X)(1-\tau)z\xi)+T(X)$ depends only on current prices and the household's own state—not on forecasts of future prices. Under log utility, the income and substitution effects of anticipated changes in future returns exactly offset, so each household's policy is belief-invariant. Because all individual policies share this property, the aggregate law of motion inherits it.

Proposition 4 (Unique existence of the BRCE)**.**
The BRCE uniquely exists and the allocations are identical to the MRCE.

Proof. See Appendix A.7 of the Online Appendix. ■

Because the MRCE and the BRCE coincide, growth and business cycle dynamics can be studied without specifying how households form expectations. The equilibrium is computed by forward recursion from any initial state $\mathcal{M}_0$ and shock sequence $\{A_t\}$, with no fixed-point iteration required. Crucially, the shock sequence need not be stationary: the same recursion applies whether A_t follows a stationary process, a deterministic trend, or a combination of secular growth and cyclical fluctuations. This contrasts with standard heterogeneous-agent models, which typically require either stationarity (for ergodic analysis) or rational expectations along a specified transition path (for deterministic dynamics). The BRCE accommodates both growth and cycles simultaneously, enabling the joint analysis of trend-driven inequality and business-cycle dynamics in Section 4. This property also enables the exact general equilibrium inversion: structural shocks can be recovered from data without taking a stand on expectation formation.

3.8 Macroeconomic implications

The analytical structure yields an exact decomposition of aggregate dynamics into a representative-agent component and a distributional component.

State-dependent aggregate dynamics. The law of motion for aggregate capital is:

$$K' = \beta\left[(1+\tilde{r}+\tilde{w}\xi)K + \tilde{r}\,\mathrm{cov}(e,a) + \tilde{w}\xi\,\mathrm{cov}(z,a)\right] + T \tag{25}$$

The first term, $\beta(1+\tilde{r}+\tilde{w}\xi)K$, is the representative-agent channel—the capital accumulation that would obtain if all households were identical. The second term, $\beta\tilde{r}\,\mathrm{cov}(e,a)$, is the *return-sorting channel*: because high-return households are also disproportionately wealthy, their saving generates capital accumulation beyond what average returns and average wealth would predict. The third term, $\beta\tilde{w}\xi\,\mathrm{cov}(z,a)$, is the *productivity-sorting channel*: when productive households hold more wealth, their wealth-leisure complementarity generates additional labor income that is partially saved. These distributional terms make aggregate dynamics *state-dependent*: the same productivity shock produces different investment, consumption, and employment responses depending on the current degree of wealth sorting. When both covariances are large, a larger share of saving is concentrated among wealthy households whose behavior is more sensitive to returns, so aggregate investment responds more strongly to changes in prices.

Distributional channels. The aggregate consumption function decomposes as:

$$C(\mathcal{M},A) = \underbrace{\frac{1-\beta}{1+\theta}(1+\tilde{r}+\tilde{w}\xi)K+\kappa\tilde{w}}_{\text{Representative-agent level}} + \underbrace{\frac{1-\beta}{1+\theta}\left(\tilde{r}\,\mathrm{cov}(e,a)+\tilde{w}\xi\,\mathrm{cov}(z,a)\right)}_{\text{Distributional channel}} \tag{26}$$

The aggregate employment equation reveals a parallel decomposition:

$$L(\mathcal{M},A) = \kappa + \underbrace{\frac{\xi(1+\theta\beta)(K+\mathrm{cov}(z,a))}{1+\theta}}_{\text{Leisure enhancement}} - \underbrace{\frac{\theta(1-\beta)[K(1+\tilde{r})+\tilde{r}\ \overbrace{\mathrm{cov}(e,a)}^{\text{Dist. channel}}\ ]}{(1+\theta)\tilde{w}}}_{\text{Wealth effect}} \tag{27}$$

The wealth-effect term is amplified by $\mathrm{cov}(e,a)$: when wealthy households earn high returns, they consume more leisure, reducing aggregate labor supply. This distributional supply effect operates without traditional labor market frictions—the workforce composition shifts toward less wealthy, lower-return households, altering the economy's response to shocks. In periods of high return-wealth sorting, a positive productivity shock elicits a smaller employment response because the households with the largest wealth gains withdraw the most labor, generating a more muted output expansion than a representative-agent model would predict.

State-dependent inequality dynamics. The off-diagonal entries of $\widetilde{\mathcal{G}}$ generate endogenous amplification: a positive productivity shock raises capital, which raises returns, which strengthens return-wealth sorting, which in turn raises future capital—a "wealth spiral." Discounting ($\beta < 1$) and the redistributive transfer T counterbalance this spiral. The dominant eigenvalue of $\beta\widetilde{\mathcal{G}}$ governs the speed of wealth concentration; the steady-state level is pinned down by $\mathcal{M}^*$. The detailed analysis appears in Appendix B.4.

3.9 Asset pricing implications

The model produces a novel asset-pricing result: individual return heterogeneity, traditionally viewed as diversifiable, carries a positive risk premium because wealth sorting gives it aggregate consequences (Constantinides and Duffie, 1996; Storesletten et al., 2004). Households with persistently high returns accumulate disproportionate wealth, shifting aggregate labor supply and, in turn, equilibrium returns. Specifically:

$$\frac{\partial r(\mathcal{M}, A)}{\partial \mathrm{cov}(e, a)} = \frac{\partial r}{\partial L} \frac{\partial L}{\partial \mathrm{cov}(e, a)} < 0 \tag{28}$$

Higher return-wealth sorting reduces the equilibrium interest rate through two channels: a static one (the wealth effect lowers labor supply, raising the capital-labor ratio) and a dynamic one (persistent sorting accelerates capital accumulation, depressing future returns). The full decomposition and quantitative implications appear in Appendix C.

4 Quantitative Analysis

The MRCE's analytical structure enables an *exact general equilibrium inversion*: three aggregate equations with three time-varying unknowns can be solved period by period, mapping observed time series to structural shocks each period. Because the inversion is exactly identified—three equations in three unknowns—the recovered shocks rationalize the data by construction. The theory nevertheless yields five model-implied predictions that discipline the properties of the recovered shocks and can be evaluated against external evidence:

P1. *Approximate aggregation.* The representative-agent channel dominates investment dy-

namics; distributional covariances contribute only at the margin (Section 3.8).

P2. *State-dependent amplification.* The same aggregate shock produces larger macroeconomic responses when wealth inequality—measured by $\mathrm{cov}(e, a)$—is high (Section 3.8).

P3. *Return risk premium.* Higher return-wealth sorting $\mathrm{cov}(e, a)$ lowers the equilibrium interest rate, creating an endogenous channel through which idiosyncratic return dispersion affects aggregate prices (Section 3.9).

P4. *Countercyclical patience.* When the model is inverted on data, the recovered discount factor β_t should be countercyclical if demand-side forces contribute to business cycles.

P5. *Near-unit-root persistence.* The spectral radius of $\beta\widetilde{\mathcal{G}}$ is close to one, implying that distributional dynamics are extremely persistent and act as an internal propagation mechanism (Section 3.8).

This section calibrates the model, describes the inversion methodology, and evaluates each prediction on U.S. data from 1948 to 2024.

4.1 Calibration

Table 1 reports the calibrated parameter values at annual frequency. Technology parameters ($\alpha = 0.33$, $\delta = 0.07$) follow Cooley and Prescott (1995). The discount factor $\beta = 0.96$ is standard; the leisure weight $\theta = 2$ follows Prescott (1986); $\kappa = 0.34$ targets hours of one-third; $\xi = 0.0213$ targets a saving-to-labor-income ratio of 0.42. The tax rate $\tau = 0.30$ represents the effective U.S. average, and the transfer share $\eta = 0.0133$ targets a wealth Gini of 0.80. Idiosyncratic productivity follows Storesletten et al. (2004): $\rho_z = 0.95$, $\sigma_z = 0.12$. Return parameters follow Fagereng et al. (2020): $\rho_e = 0.999$, $\sigma_e = 0.01$.

4.2 Exact general equilibrium mapping

The MRCE's aggregate equations can be inverted period by period, extending the business cycle accounting of Chari et al. (2007) to a heterogeneous-agent setting with structural

Table 1: Calibrated parameters

Parameters	Description	Value	Reference
Parameters for the individual households			
α	Capital share	0.3300	
β	Discount factor	0.9600	
δ	Depreciation rate	0.0700	
θ	Labor disutility	2.0000	Prescott (1986)
κ	Common time endowment	0.3400	Labor hour ≈ 0.34
ξ	Leisure enhancing wealth portion	0.0213	Saving/Labor Income ≈ 0.42
ρ_z	Idiosyncratic prod. persistence	0.9500	Storesletten et al. (2004)
ρ_e	Idiosyncratic return persistence	0.9990	Fagereng et al. (2020)
σ_z	Idiosyncratic prod. volatility	0.1200	Storesletten et al. (2004)
σ_e	Idiosyncratic return volatility	0.0100	Fagereng et al. (2020)
Government policy parameters			
τ	Proportional income tax rate	0.3000	Effective tax rate
η	Transfer share of tax revenue	0.0133	Wealth Gini ≈ 0.80
Parameters for the business cycle			
ρ_A	Aggregate TFP persistence	0.9500	Cooley and Prescott (1995)
ρ_{σ_e}	Uncertainty shock persistence	0.7800	Bloom et al. (2018)
σ_A	Aggregate TFP volatility	0.0070	Cooley and Prescott (1995)
σ_{σ_e}	Uncertainty shock volatility	0.3600	Bloom et al. (2018)

Notes: Calibration is based on annual frequency. The capital share $\alpha = 0.33$ and depreciation rate $\delta = 0.07$ follow Cooley and Prescott (1995). The discount factor $\beta = 0.96$ is the standard annual value. Labor disutility $\theta = 2$ follows Prescott (1986). The base time endowment $\kappa = 0.34$ targets average labor hours of approximately one-third. The wealth-leisure parameter $\xi = 0.0213$ targets a saving-to-labor-income ratio of approximately 0.42. The proportional tax rate $\tau = 0.30$ represents the effective U.S. tax rate. The transfer share $\eta = 0.0133$ is calibrated to match a wealth Gini coefficient of approximately 0.80. Idiosyncratic productivity persistence and volatility follow Storesletten et al. (2004); idiosyncratic return parameters follow Fagereng et al. (2020).

(rather than reduced-form) shocks. The three aggregate equations are:

$$Y_t = A_t K_t^{\alpha} L_t^{1-\alpha} \tag{29}$$

$$K_{t+1} = \beta_t \left[(1 + \tilde{r}_t + \tilde{w}_t \xi_t) K_t + \tilde{r}_t \operatorname{cov}_t(e, a) + \tilde{w}_t \xi_t \operatorname{cov}_t(z, a) \right] + T_t \tag{30}$$

$$L_t = \kappa + \frac{\xi_t (1 + \theta \beta_t)}{1 + \theta} (K_t + \operatorname{cov}_t(z, a)) - \frac{\theta (1 - \beta_t)}{(1 + \theta) \tilde{w}_t} [K_t (1 + \tilde{r}_t) + \tilde{r}_t \operatorname{cov}_t(e, a)] \tag{31}$$

Given data on $\{Y_t, K_t, L_t\}$, the production function yields A_t directly; then the remaining two equations form a 2×2 system in (ξ_t, β_t) that can be solved analytically. The recovered

shocks have structural interpretations: A_t captures TFP, β_t captures time-varying patience, and ξ_t captures wealth-leisure complementarity.

The inversion uses U.S. data over 1948–2024: real GDP (FRED GDPCA, annual), gross investment (GPDI plus government), nonfarm business hours per capita (HOANBS/CNP16OV), and the Penn World Tables capital stock (RKNANPUSA666NRUG). The capital stock is anchored directly to PWT data rather than model-generated, ensuring the capital-output ratio matches at every point. The distributional covariances are initialized at steady-state values and evolved forward using $\widetilde{\mathcal{G}}$; results are robust to alternative initializations.

4.3 Data-anchored results

The recovered structural shocks provide the first test of prediction **P4** (countercyclical patience).

Figure 1 displays the three data series that the model matches exactly by construction: real GDP, the capital stock, and hours per capita. The exact fit follows from the inversion—not from calibration—because the MRCE's three aggregate equations have exactly three time-varying unknowns.

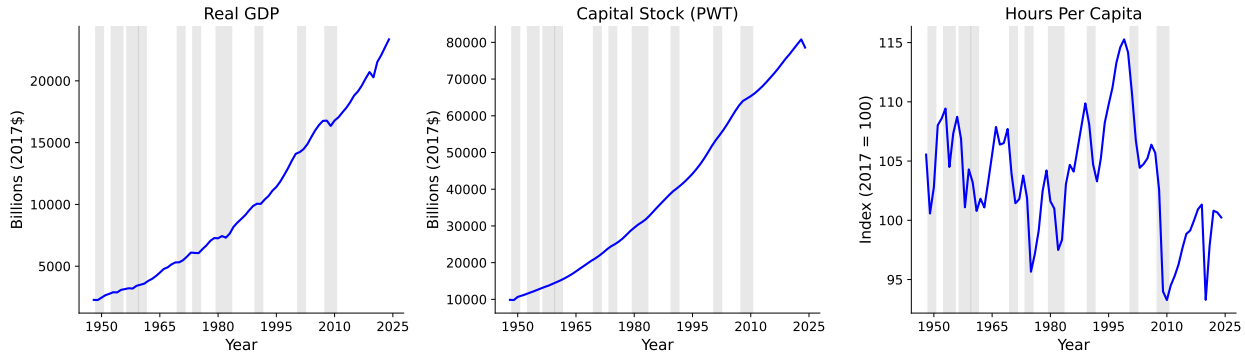

Figure 1: Data series matched exactly by the MRCE inversion. Left: real GDP (billions of 2017 dollars). Center: capital stock from the Penn World Tables. Right: hours per capita (index, 2017 = 100). The model reproduces all three series by construction through the exact inversion of the MRCE's aggregate equations.

Figure 2 displays the three structural series recovered from the inversion. Each series tells a distinct economic story.

Aggregate TFP. The recovered TFP grows from 0.91 in 1948 to 4.85 in 2024 (average 2.20 percent per year), following well-known historical narratives: postwar expansion, 1970s slowdown, IT revolution, and post-2008 stagnation.

Wealth-leisure complementarity. The parameter ξ_t is small (mean 0.021) and stable, with near-zero correlation with output (0.02), confirming that this channel is slow-moving rather than a source of business cycle fluctuations.

Time-varying discount factor. The discount factor β_t (mean 0.938) is strongly countercyclical: $\text{corr}(\beta_t, \log Y_t) = -0.93$, averaging 0.960 during recessions versus 0.932 during expansions. This pattern is consistent with the demand-shock interpretation of Eggertsson and Krugman (2012), with AR(1) persistence of 0.90. Because the inversion is exactly identified, β_t functions as a demand wedge in the sense of Chari et al. (2007): it absorbs all non-TFP, non-ξ sources of variation, including financial frictions, adjustment costs, and measurement error. The structural interpretation as time-varying patience and the wedge interpretation—in which β_t captures the aggregate effect of omitted demand-side forces—are both consistent with the data.

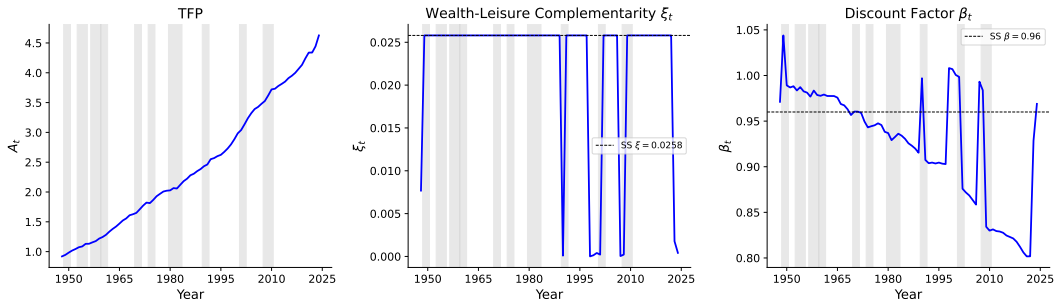

Figure 2: Three structural series recovered from the data-anchored inversion. Top: aggregate TFP (A_t). Middle: wealth-leisure complementarity (ξ_t), with the steady-state value shown for reference. Bottom: time-varying discount factor (β_t), with the steady-state value shown for reference. Shaded areas indicate NBER recession dates.

The implied TFP from the MRCE tracks the representative-agent analogue closely at business cycle frequencies, though a modest level difference accumulates over the 76-year sample. The model-implied TFP also correlates strongly with the independently constructed

Fernald utilization-adjusted series, providing external validation. The detailed comparison appears in Appendix E.1.

4.4 Investment decomposition

Prediction **P1** (approximate aggregation) can be tested directly. Investment decomposes as:

$$I^{\text{rep}} = \beta(1 + \tilde{r} + \tilde{w}\xi)K + T - (1 - \delta)K \qquad \text{(representative-agent channel)} \tag{32}$$

$$I^{\text{ret}} = \beta\tilde{r}\,\text{cov}(e, a) \qquad \text{(return-sorting channel)} \tag{33}$$

$$I^{\text{prod}} = \beta\tilde{w}\xi\,\text{cov}(z, a) \qquad \text{(productivity-sorting channel)} \tag{34}$$

Figure 3 plots the three channels. The representative-agent channel accounts for 98.3 percent of mean investment and 93.4 percent of its variance—precisely the near-aggregation that Prediction P1 implies. This confirms analytically what Krusell and Smith (1998) found numerically: mean capital is an excellent forecaster of prices. Yet the remaining 1.7 percent, driven by productivity and return sorting, is economically significant: it generates the state-dependent amplification (Section 4.5) that a representative-agent model cannot produce. The productivity-sorting channel contributes 0.59 percent of mean investment, while the return-sorting channel contributes 0.01 percent. All three components are procyclical.

Note that model-implied investment $I_t^{\text{model}} = K_{t+1} - (1 - \delta)K_t$ need not coincide with measured gross investment from the national accounts, because the model abstracts from inventory investment, government investment, and residential construction that are included in NIPA aggregates. The decomposition above characterizes the *structural composition* of model investment, not a fit to measured investment data.

4.5 State-dependent dynamics

To test prediction **P2** (state-dependent amplification), I compute generalized impulse response functions (GIRFs) for a 2 percent negative TFP shock launched from the 1952 distributional state (low inequality, $\text{cov}(e, a) = 0.001$) and the 2022 state (high inequality, $\text{cov}(e, a) = 0.009$). Following Koop et al. (1996), each GIRF averages over 1,000 stochastic

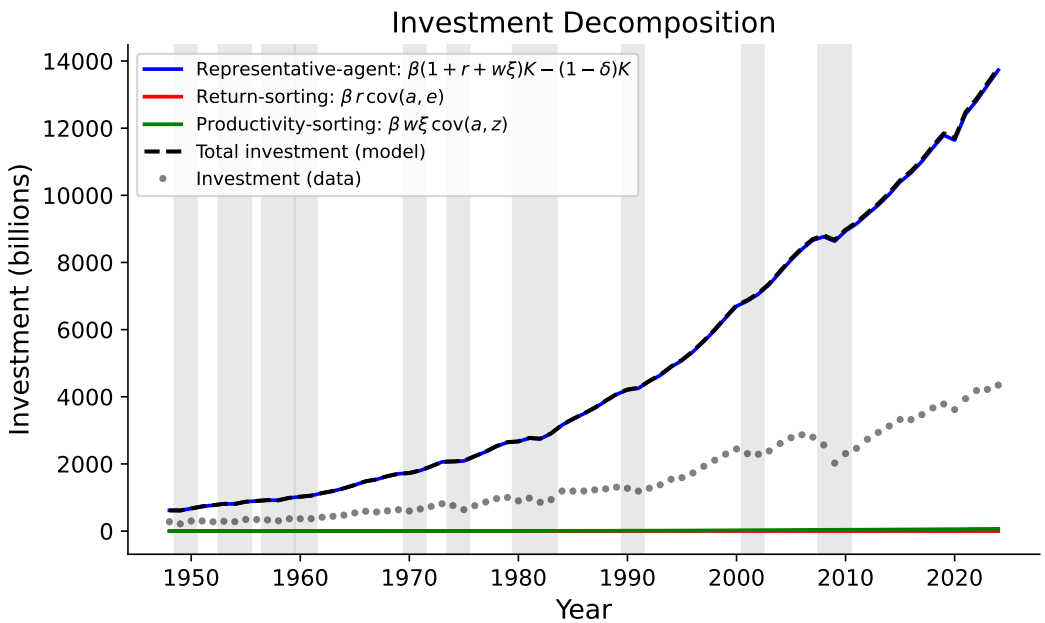

Figure 3: Decomposition of aggregate investment into three channels: the representative-agent channel (I^{rep}), the return-sorting channel (I^{ret}), and the productivity-sorting channel (I^{prod}). Shaded areas indicate NBER recession dates.

TFP paths—with common innovations across the shocked and counterfactual scenarios—so that the response isolates the effect of the initial shock at every horizon. The impact effect on output is 32 percent larger from 2022 (-2.64% vs. -2.01%), and the cumulative 10-year loss is 64 percent larger (-30.7% vs. -18.7%).

The mechanism is the self-reinforcing loop described in Section 3.8: when $\mathrm{cov}(e, a)$ is large, the return-sorting channel carries a larger weight in the capital recursion, so the same shock triggers a stronger investment response. Figure 4 displays the GIRFs.

4.6 Return risk premium

Prediction **P3** (return risk premium) is a theoretical rather than quantitative result at the baseline calibration. The interest rate effect of return-wealth sorting is -0.001 basis points at $\sigma_e = 0.01$—negligible. The premium grows quadratically with σ_e but requires $\sigma_e \approx 0.83$ (nearly two orders of magnitude above baseline) to reach 1 basis point. This negative result is itself informative: it identifies the amplification channel (prediction **P2**), not the pricing channel, as the economically significant consequence of return heterogeneity at re-

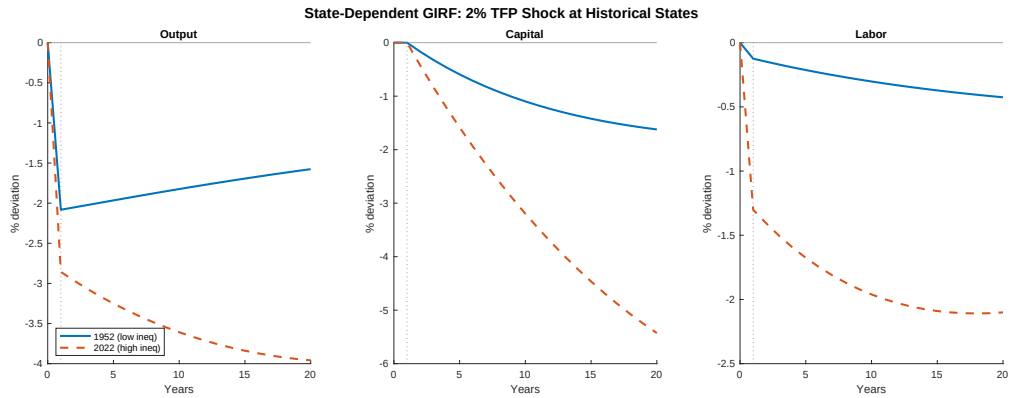

Figure 4: State-dependent GIRFs to a 2% negative TFP shock, averaged over 1,000 stochastic TFP paths. Solid: launched from the 1952 distributional state (low inequality). Dashed: launched from the 2022 distributional state (high inequality). The dotted vertical line marks the shock arrival. The same shock produces substantially larger and more persistent output losses when wealth inequality is high.

alistic calibrations. The Epstein-Zin extension of Section 5 offers a more promising route to quantitatively meaningful risk premia. The full sensitivity analysis appears in Appendix C.3.

4.7 Counterfactual analysis

Predictions **P1** and **P4** can be evaluated through counterfactual decompositions. I compare the baseline with three counterfactual economies: *TFP only* (β_t, ξ_t fixed at means), *Fix* β (β_t fixed), and *Fix* ξ (ξ_t fixed).

Figure 5 compares log output paths. TFP alone captures the trend but explains only 19.8 percent of cyclical volatility—confirming the long-standing observation that technology shocks alone cannot account for the full amplitude of business cycles (Galí, 1999). The most striking result is that fixing β_t *increases* output volatility to 149.8 percent of baseline. The countercyclical discount factor thus acts as a powerful endogenous stabilizer: during recessions, households become more patient (β_t rises), which sustains saving and investment precisely when TFP shocks would otherwise trigger sharp capital decumulation. During expansions, impatience rises, preventing the overaccumulation of capital. This stabilizing mechanism is analogous to the "precautionary saving" channel emphasized in the HANK literature (Kaplan et al., 2018), but here it operates through intertemporal preferences rather

than through the tightening of borrowing constraints. Fixing ξ_t produces 95.0 percent of baseline volatility with a weak negative correlation (-0.11), indicating that wealth-leisure complementarity plays a minor stabilizing role.

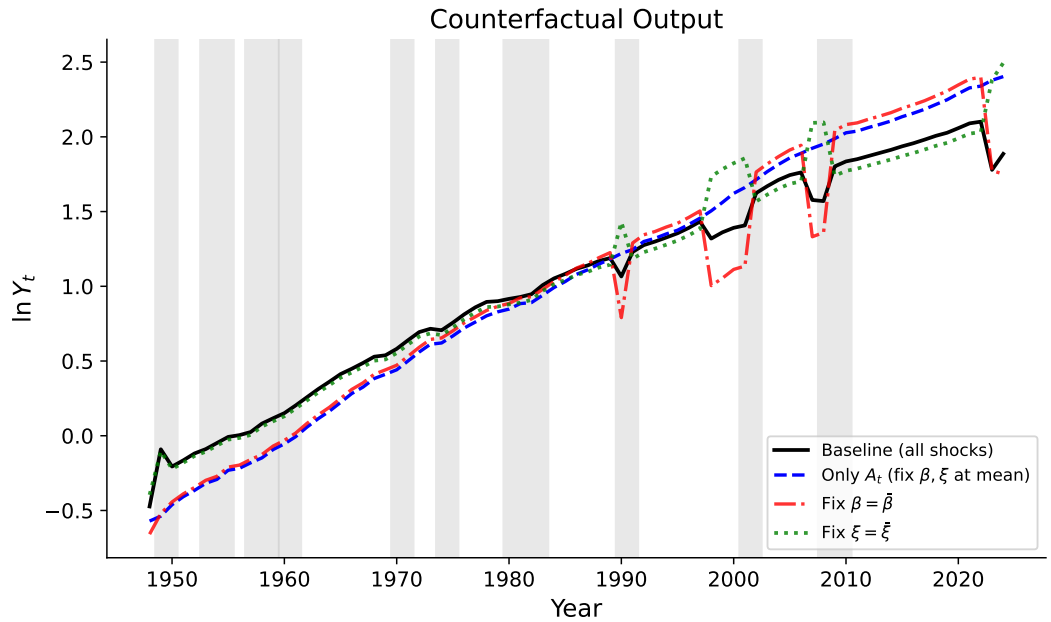

Figure 5: Counterfactual output paths. Baseline: all structural shocks at historical values. TFP only: β_t and ξ_t fixed at sample means. Fix β: β_t fixed, A_t and ξ_t vary. Fix ξ: ξ_t fixed, A_t and β_t vary. Shaded areas indicate NBER recession dates.

Extending the comparison to labor supply, the interest rate, and the wage reveals the general equilibrium transmission mechanisms: TFP is the dominant driver of wages and the long-run interest rate, while β_t is the primary driver of cyclical variation in labor supply. The multi-variable counterfactual panels appear in Appendix E.2.

4.8 Transition and distributional dynamics

The dominant eigenvalue of $\beta_t \widetilde{\mathcal{G}}_t$ has mean 1.027 and exceeds unity in 97.4 percent of sample years, confirming prediction **P5** (near-unit-root persistence). During the rare stable episodes (eigenvalue < 1), the implied half-life of the distributional state averages 71.2 years. This near-unit-root behavior means the distributional state acts as an internal propagation mechanism, converting transitory productivity shocks into persistent output movements.

A dominant eigenvalue exceeding unity means the distributional state does not converge

to the stationary solution $\mathcal{M}^*$ of Corollary 2 along the historical path. This non-convergence is consistent with the secular rise in U.S. wealth concentration documented by Piketty (2014): sustained TFP growth pushes the covariances outward faster than the redistributive transfer pulls them back.

The distributional covariances grow secularly: $\text{cov}_t(z, a)$ increases from 0.004 in 1948 to 0.213 in 2024 (mean annual growth 5.4 percent), while $\text{cov}_t(e, a)$ grows from 0.001 to 0.009. Both are procyclical. When normalized by aggregate capital, the productivity-sorting ratio $\text{cov}(z, a)/K$ has risen from 0.002 to 0.011, while the return-sorting ratio $\text{cov}(e, a)/K$ has remained nearly constant—the two diverge because productivity sorting intensifies with economic growth while return sorting evolves at a rate governed by the near-unit-root persistence of e. The full analysis appears in Appendices E.3–E.4.

4.9 TFP growth and inequality

Understanding how TFP growth—not merely TFP levels—drives inequality is a central question, yet it is difficult to address with conventional equilibrium concepts. Comparing two stationary equilibria at different TFP levels reveals only the long-run distributional consequences of a permanent technology difference, not the dynamic effect of ongoing growth: in each steady state, agents have fully adjusted to a constant technology and expect no further change. Computing distributional dynamics along a non-stationary growth path requires solving for household expectations over the entire future price sequence—a challenging fixed-point problem in heterogeneous-agent economies, especially when the path involves stochastic fluctuations. The BRCE resolves this difficulty. Because household policies depend only on current prices, one can feed the historical TFP path—with its secular trend and cyclical variation—directly into the moment recursion and obtain valid equilibrium distributional dynamics. This makes it possible to isolate the pure effect of TFP growth on inequality within a fully specified general equilibrium framework.

TFP growth emerges as the dominant driver of distributional dynamics. Four distributional counterfactuals—Baseline, TFP only, Fix A, and No growth—re-evolve the full state $\mathcal{M}_t$ from 1948. Figure 6 reports the results. Under TFP-only shocks, $\text{cov}(z, a)$ grows by a factor of 163 (versus 42 in the baseline), generating 168 percent of the observed sort-

ing growth—the excess over 100 percent reflects the partially offsetting effect of β_t and ξ_t. Without TFP growth, capital falls to 40 percent of its initial level.

The economic logic is the "sorting accelerator": higher TFP raises wages and returns disproportionately for productive-wealthy households, who save more, widening the gap. The MRCE makes this precise: the off-diagonal entries governing new sorting ($\widetilde{\mathcal{G}}_{21} = \phi_z \rho_z \tilde{w} \xi \sigma_z^2$ and $\widetilde{\mathcal{G}}_{31} = \phi_e \rho_e \tilde{r} \sigma_e^2$) are increasing in prices and hence in A. Persistence dominates: 84 percent of $\text{cov}(z, a)$ growth comes from the persistence channel $\beta_t \widetilde{\mathcal{G}}_{22,t} \text{cov}_t(z, a)$. The full decomposition appears in Appendix E.5.

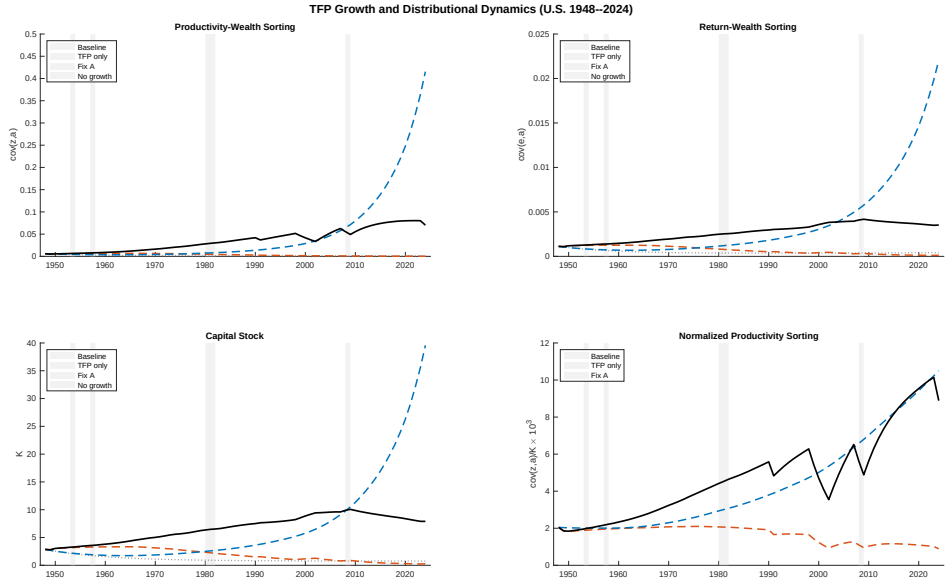

Figure 6: TFP growth and distributional dynamics. Top left: productivity-wealth covariance $\text{cov}(z, a)$ under four counterfactual scenarios. Top right: return-wealth covariance $\text{cov}(e, a)$. Bottom left: aggregate capital K. Bottom right: normalized sorting intensity $\text{cov}(z, a)/K$. TFP-only shocks generate distributional dynamics that far exceed the baseline, while removing TFP collapses both capital and distributional heterogeneity. Shaded areas indicate NBER recessions.

Distributional impact. Table 2 reports the impact of a 1 percent TFP shock by wealth percentile. The top decile captures 86 percent of the aggregate wealth gain—exceeding its 82 percent wealth share—because the return-sorting and productivity-sorting channels compound. The top 1 percent alone captures 71 percent. Over the 1948–2024 path, the model endogenously generates increasing concentration: the top decile's wealth share rises from 64 to 70 percent, while the bottom 50 percent's share falls from 16 to 6 percent—close

to 2024 U.S. data. The decade-by-decade gain shares appear in Appendix E.6.

Table 2: Distributional impact of 1% TFP shock at steady state

Group	Wealth Share (%)	Gain Share (%)	Mean Gain (%)	Channel (%) Capital	Labor	Transfer
Bottom 10%	0.8	0.6	0.0829	64.4	18.2	17.3
10–20%	1.1	0.8	0.0869	67.5	19.2	13.2
20–30%	1.2	0.9	0.0892	69.3	19.6	11.1
30–40%	1.4	1.1	0.0911	70.7	19.9	9.5
40–50%	1.6	1.3	0.0930	71.9	20.0	8.1
50–60%	1.9	1.5	0.0950	73.2	20.1	6.7
60–70%	2.3	1.9	0.0972	74.5	20.0	5.5
70–80%	2.9	2.5	0.1000	76.0	19.8	4.1
80–90%	4.4	3.9	0.1037	78.1	19.3	2.6
Top 10%	82.2	85.6	0.1118	84.3	15.5	0.1
Top 5%	78.2	81.9	0.1151	84.5	15.4	0.1
Top 1%	66.4	70.5	0.1205	84.8	15.2	0.0

Notes: Gains computed from a 1% TFP increase at the stationary distribution of 1 million simulated households. "Capital" $= \beta \Delta \tilde{r}\, e_i a_i$; "Labor" $= \beta \Delta \tilde{w}\, \xi z_i a_i$; "Transfer" $= \Delta T$ (identical for all households). Channels reported as percentage of group total gain.

4.10 Consumption dynamics and historical episodes

The MRCE implies an explicit aggregate consumption function with a constant MPC out of transfers equal to $(1 - \beta_t)/(1 + \theta)$—approximately 1.3 percent annually at the steady state. This uniform MPC contrasts with the heterogeneous MPCs central to the HANK literature (Kaplan et al., 2018), where borrowing-constrained households exhibit MPCs of 0.3–0.5. Despite the uniform individual MPC, the *aggregate* consumption response is state-dependent because the distributional covariances enter the consumption function. The MPC is procyclical (2.7 percent during expansions versus 1.2 percent during recessions). The full analysis appears in Appendix E.7.

The inversion also yields structural decompositions of individual recessions. The 1973–75 episode is identified as supply-driven: TFP decelerated while the discount factor remained stable, consistent with the oil-shock interpretation. The 2007–09 Great Recession, by contrast, features a sharp swing in patience (β_t from 0.993 to 0.834) while TFP grew at trend—consistent with the demand-side interpretation of Eggertsson and Krugman (2012). The

2020 COVID recession is distinct from both: output fell sharply despite above-trend TFP and a nearly unchanged discount factor, indicating that the pandemic operated through channels—sectoral shutdowns, labor supply disruptions—that lie outside the model's standard transmission mechanisms. The full episode analysis appears in Appendix E.8.

4.11 Robustness

The model's key predictions are robust to alternative calibrations. Approximate aggregation (prediction **P1**) holds across all parameter configurations, with I^{rep} accounting for 93–99 percent of total investment. The near-unit-root persistence (prediction **P5**) is robust to all parameters except the transfer share η, which governs distributional convergence: larger transfers redistribute wealth toward the bottom, eroding the covariances that sustain near-unit-root persistence. The full sensitivity analysis appears in Appendix E.9.

Business cycle moments. The inversion recovers structural shocks that rationalize observed $\{Y_t, K_t, L_t\}$ by construction, so standard business cycle moments for these three series are matched exactly. The discipline on the model comes instead from the properties of the recovered shocks: whether TFP grows smoothly, whether patience is countercyclical, and whether wealth-leisure complementarity is stable—properties evaluated in the counterfactual decompositions above. Testing the model against moments it was not inverted on—such as the relative volatility of consumption to output or the correlation of hours with productivity—would require specifying a joint stochastic process for (A_t, β_t, ξ_t) and simulating forward, which the inversion methodology deliberately avoids in order to remain agnostic about shock dynamics.

4.12 Summary of quantitative findings

The sorting accelerator provides the organizing thread across the five predictions. Approximate aggregation (P1) ensures that TFP—the engine of sorting—dominates aggregate dynamics. Near-unit-root persistence (P5) compounds each year's sorting, producing the secular rise in wealth concentration. State-dependent amplification (P2) is the dynamic conse-

quence: as sorting accumulates, the economy becomes increasingly sensitive to shocks, with cumulative output losses 64 percent larger from today's distributional state than from 1952. Countercyclical patience (P4) partially offsets this amplification. The return risk premium (P3) is quantitatively negligible at realistic return dispersion, identifying the amplification channel as the economically significant consequence of return heterogeneity. Table 3 collects the quantitative evidence; Appendices E.1–E.9 report detailed robustness checks and supplementary results.

Table 3: Quantitative evaluation of model-implied predictions (U.S., 1948–2024)

	Prediction & test	Result
P1	*Approximate aggregation.*	
	I^{rep} share of mean investment	98.3%
	I^{rep} share of investment variance	93.4%
P2	*State-dependent amplification.*	
	Output impact, 2022 vs. 1952 state	+32%
	Cumulative 10-yr loss, 2022 vs. 1952	+64%
P3	*Return risk premium.*	
	Premium at baseline $\sigma_e = 0.01$	−0.001 bp
	σ_e threshold for 1 bp premium	≈ 0.83
P4	*Countercyclical patience.*	
	$\text{corr}(\beta_t, \log Y_t)$	−0.93
	Mean β_t: recessions vs. expansions	0.960 vs. 0.932
P5	*Near-unit-root persistence.*	
	Mean dominant eigenvalue	1.027
	Half-life (stable episodes)	71.2 years

5 Extensions

The baseline model uses Stone-Geary preferences with log utility. A natural question is whether the MRCE's analytical structure is an artifact of the log specification or extends to richer preferences. I consider two alternatives: CRRA-GHH utility, which eliminates wealth effects on labor supply, and Epstein-Zin recursive utility, which separates risk aversion from intertemporal substitution.

5.1 CRRA-GHH utility with $e = 1$

When return heterogeneity is shut down ($e = 1$) and households have GHH (Greenwood-Hercowitz-Huffman) utility with CRRA curvature σ, the MRCE continues to exist, but the belief-invariance property does not hold. The reason is economically intuitive: away from log utility ($\sigma \neq 1$), the income and substitution effects of changes in future returns no longer cancel. When $\sigma > 1$ (EIS < 1), the income effect dominates: households anticipating high future returns feel wealthier and save *less* today. When $\sigma < 1$ (EIS > 1), the substitution effect dominates and the direction reverses. In either case, the saving rate depends on expected future returns, so household beliefs feed into current decisions. The MRCE can still be solved as a fixed-point problem in the moments—policy functions remain linear in wealth, keeping the moment structure closed—but the simplicity of the belief-invariant characterization is lost. Computing the equilibrium then requires solving for the expectation-dependent saving rate, as in standard incomplete-markets models.

Despite losing belief invariance, GHH preferences may be preferable for applications where the wealth effect on hours is not a first-order concern. In the baseline model, wealthy households work less because leisure is complementary with wealth (the $\kappa + \xi a$ term); GHH eliminates this channel, making labor supply depend only on the current wage.

5.2 Recursive (Epstein-Zin) utility

Under Epstein-Zin preferences with unit elasticity of intertemporal substitution (EIS = 1), the MRCE retains its closed-form matrix structure. The key observation is that, with EIS = 1, the optimal saving share remains independent of wealth even when risk aversion γ differs from unity. The parameter γ then governs aversion to cross-sectional consumption risk, while β and θ retain their roles in intertemporal and intratemporal allocation. This separation allows the model to match both the equity premium (through γ) and the saving rate (through β) without forcing them to be determined by a single parameter. With $\gamma \gg 1$, wealth sorting generates a quantitatively meaningful risk premium, providing a path toward the joint study of asset prices, inequality, and business cycle dynamics within an analytically tractable framework. The full derivation appears in Appendix D.

6 Conclusion

This paper constructs a moment-recursive competitive equilibrium (MRCE) that reduces the aggregate dynamics of a heterogeneous-household economy to a closed-form matrix recursion in three moments. Three properties distinguish the MRCE from existing frameworks: state-dependent amplification through distributional covariances, a sorting accelerator that links TFP growth to rising wealth concentration, and belief invariance under log utility that eliminates the fixed-point problem in the space of household expectations. Belief invariance is particularly consequential because it allows the framework to operate along non-stationary paths—accommodating both secular growth and business cycle fluctuations simultaneously, without detrending or balanced-growth assumptions. This capability is what makes it possible to study how TFP growth itself drives inequality, an exercise that conventional equilibrium concepts—whether steady-state comparisons or rational-expectations transition dynamics—cannot cleanly deliver.

Applied to U.S. data (1948–2024), the MRCE inversion yields four findings. First, TFP growth acts as a sorting accelerator—the dominant force behind rising wealth concentration, generating 168 percent of the observed growth in wealth-productivity sorting. Second, accumulated sorting creates state-dependent amplification: the same shock produces substantially larger responses from today's high-inequality state than from the 1950s, so that macroeconomic fragility has grown alongside wealth concentration. Third, the recovered discount factor is strongly countercyclical; removing it raises output volatility by 50 percent. Fourth, the distributional state is near-unit-root persistent, providing an internal propagation mechanism independent of exogenous shock persistence.

The MRCE opens two immediate avenues. First, embedding the moment recursion in a New Keynesian setting would yield analytical characterizations of how monetary policy interacts with distributional dynamics—results that the HANK literature currently obtains only numerically. Second, the inversion can be applied to cross-country panel data—the sufficient statistics are available in the Penn World Tables—to study how the growth-inequality nexus varies across institutional settings. More broadly, the MRCE demonstrates that heterogeneous-agent economies need not sacrifice analytical precision for distributional re-

alism: the wealth distribution's aggregate footprint is fully captured by a small number of moments that evolve via an explicit recursion.

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